

I1 extends I2

Program Code:

```
package core_java;

interface I2 {
    void connect();
}

interface I1 extends I2 {
    void sendData();
}

class C1 implements I1 {

    public void connect() {
        System.out.println("Network connected");
    }

    public void sendData() {
        System.out.println("Data sent successfully");
    }

    void disconnect() {
        System.out.println("Network disconnected");
    }
}

public class I1toI2 {
    public static void main(String[] args) {

        C1 obj = new C1();

        obj.connect();
        obj.sendData();
        obj.disconnect();
    }
}
```

Explanation:

This program demonstrates interface inheritance. Interface I1 extends interface I2, so I1 inherits the connect() method from I2 and also declares sendData(). Class C1 implements I1 and therefore provides implementations for both interface methods. Class C1 also contains its own disconnect() method.

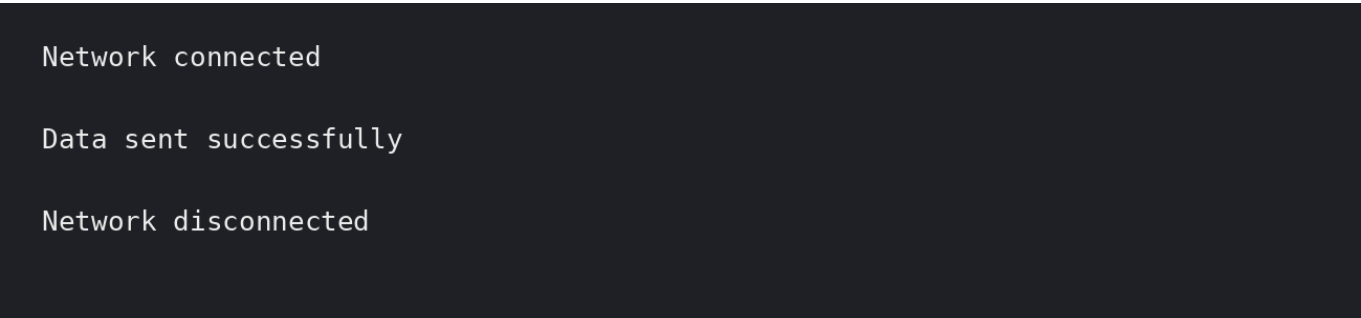
Compilation and Execution Commands:

Save the file as: I1toI2.java

Compile:
javac -d . I1toI2.java

Run:
java core_java.I1toI2

Output Screenshot:



```
Network connected

Data sent successfully

Network disconnected
```

Verification:

The output confirms that C1 successfully implements both the inherited connect() method from I2 and the sendData() method declared in I1.

Result:

Thus, the program was successfully executed and the interface inheritance relationship I1 extends I2 was verified successfully.